

Success At One Pitch: Prediction Model for Batting and Pitching

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Abstract

This paper presents the development of multiple multilinear regression models designed to predict the success rate of a batter and pitcher in a single pitch. We call this statistic ‘success-at-one-pitch’ percentage. For batters, a good outcome is hit into play and does not lead to an out, whereas a bad outcome is a strikeout or hit into play and leads to an out—and vice versa for pitchers. These models can be used on MLB data to understand a player’s success in a single pitch. By analyzing various performance metrics, these models provide insights into player evaluations.

Acknowledgments

The sources below are used to this project:

- Link to publicly accessible data by Baseball Savant: <https://baseballsavant.mlb.com/>
- Used to define our metrics, “Everything to know about Statcast’s new bat tracking data”: <https://www.mlb.com/news/what-you-need-to-know-about-statcast-bat-tracking>

We would like to thank the Women in Sports Data Hackathon team for giving us an opportunity to participate in this event and use this data!

Introduction

Research Questions:

- Which variables matter the most/least for a batter? For a pitcher?
- To what extent does each variable for batting and pitching affect the outcome of a single pitch?

The batting variables are launch angle sweet spot percentage, exit velocity, bat speed, swing length, and barrel percentage. We chose these variables because current research and statistics demonstrate that these factors have a significant impact on batting outcomes.¹ For our pitching variables—spin rate, speed, x-movement, and z-movement, we chose them because they are common metrics for pitching and were accessible in our dataset.

Important Definitions for Batting

- Success percent (success-at-one-pitch percentage): likelihood of not striking out nor having a play that results in an out over all outcomes

¹“Everything to know about Statcast’s new bat tracking data”: <https://www.mlb.com/news/what-you-need-to-know-about-statcast-bat-tracking>

- Moment of contact: time of the first local minima of the sum of XYZ displacement of the ball from home plate
- Launch angle sweet spot percentage: the fraction of time that a batter hits the ball in the launch angle sweet spot, defined as between 8 to 32 degrees.²
- Exit velocity (mph): the instantaneous speed of the ball at the moment of contact
- Bat speed (mph): the speed of the bat at the moment of contact
- Swing length (feet): the average total distance in XYZ-space the bat traveled during the swing, from the start of bat tracking data (150 ms) to the moment of contact³
- Barrel percentage: the fraction of time that a batter hits a barrel. A barrel is defined as a batted ball with an exit velocity of greater than 98 mph and within a certain range of launch angles.⁴

Important Definitions for Pitching

- Success percent (success-at-one-pitch percentage): likelihood of throwing a strike or having the play result in an out
- Spin rate (rpm): how fast the ball is spinning
- Speed (mph): how fast the ball is traveling
- X-movement (feet): the difference between minimum and maximum x-position of the ball from its sample positions before crossing home plate
- Z-movement (feet): the difference between minimum and maximum z-position of the ball from its sample positions before crossing home plate

The Batting Model

We used five variables to create the initial batting model which were launch angle sweet spot percentage, average exit velocity, average bat speed, average swing length, and average barrel percentage. Figure 1 demonstrates the collinearity between variables. We calculated some multicollinearity between variables, however, our model prioritizes low p -values over low VIF values.

We used backward selection to find the variables with p -values less than 0.1 They were launch angle sweet spot percentage (x_1), average swing length (x_2), and average exit velocity (x_3). The following equation is our model:

$$\text{Batting Success \%} = 0.5616x_1 + (-2.6716x_2) + 0.3162x_3 \text{ (R}^2\text{adj} = 0.474)$$

We made predictions using the three hitters who had the most pitches thrown to them in our dataset. The more bats a player has, the more accurately our model makes predictions.

²Sweet spot full definition: <https://www.mlb.com/glossary/statcast/sweet-spot>

³Swing length full definition: <https://www.mlb.com/glossary/statcast/swing-length>

⁴Barrel full definition: <https://www.mlb.com/glossary/statcast/barrel>

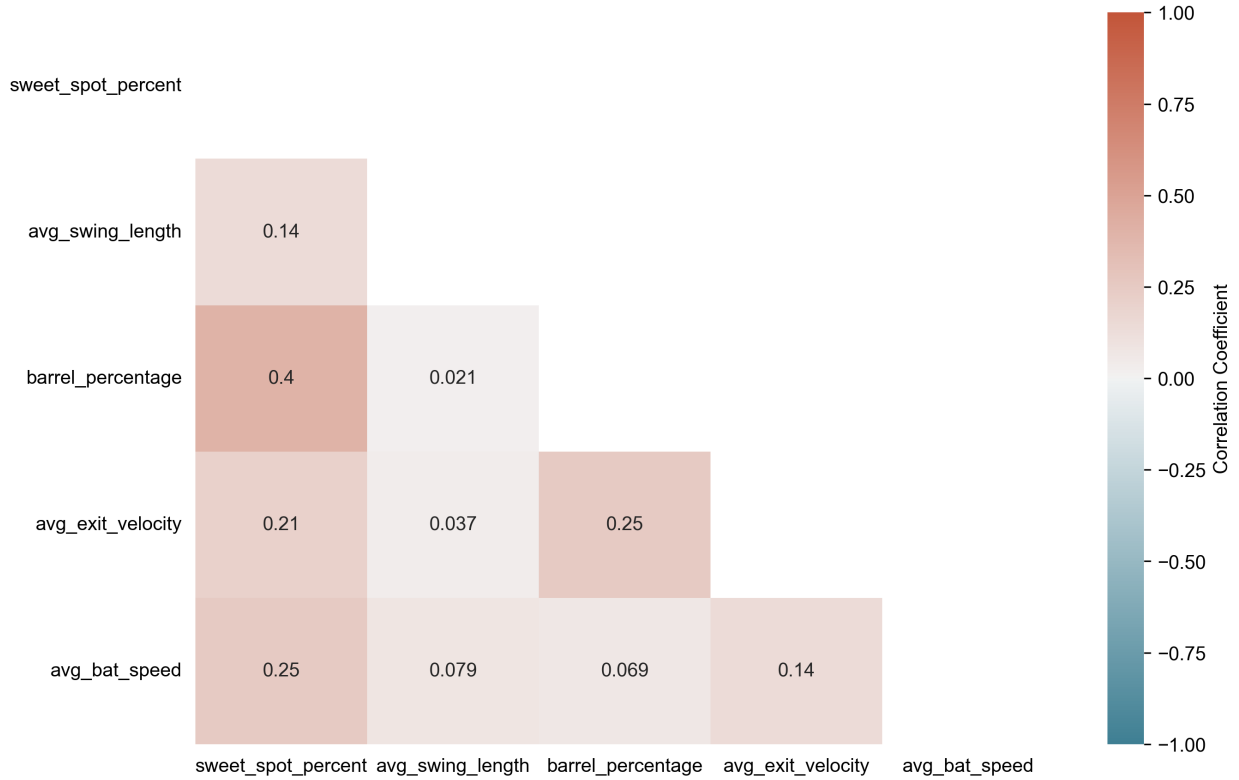


Figure 1: Correlation Coefficients of Predictors in Batting

We also used our model on three MLB hitters with the highest batting averages in the 2024 season. See full table in Appendix.

Batter ID	Number of Pitches	Success % (Data)	Success % (Model)
797957728	23	0.217	0.271
485007791	20	0.200	0.28
352830460	18	0.167	0.189

Table 1: Comparison of Success Rates: Batters with Highest Success Rates in Data

MLB Player	Batting Average	Success %
Bobby Witt Jr.	0.344	0.329
Steven Kwan	0.341	0.342
Shohei Ohtani	0.312	0.335

Table 2: MLB Players Predicted Success Rate by Pitch

From Table 1 we can see that the success rates are similar in our data and as predicted by our model. In Table 2 we see that the success rate of the MLB players is similar to their batting average. Across the minor league and MLB players, we see that the MLB players

have a higher success rate as expected and that the minor league players perform about 10% less.

The Pitching Model

We used four variables to create the initial pitching model which were spin rate, speed, x-movement, and z-movement. Since different types of pitches rely on different variables, we additionally made a model for each pitch type: Slider, Sinker, Four Seam Fastball, Cutter, Changeup, and Curveball as identified in the data. Similar to our batting model we examined multicollinearity but prioritized low p -values over low VIF values.

We found that Four Seam Fastball ($n = 33$) and Curveball ($n = 39$) pitches could not be accurately modeled as the variables had high p -values. (More data is necessary to create these models). The following table shows the selected coefficients for each pitch type model.

Pitch Type	Spin Rate (rpm)	Speed (mph)	X-Movement (feet)	Z-Movement (feet)	Number of Samples	Adjusted R-Squared
All	0	0	0.0019	0.005	1233	0.013
Slider	-0.0002	0	0	0	76	0.024
Sinker	0	0.0501	0	0	106	0.041
Cutter	0	0	-0.0353	0.1204	24	0.142
Changeup	0	0.1152	-0.1694	0	31	0.137

Table 3: Pitch Type Models

For all pitches, more movement in the x and z direction increases success rate. A slower spin rate for sliders makes for a greater success percentage. Speed matters the most for sinkers. Cutter have higher success with less x-movement and more z-movement. Throwing faster changeup pitches with reduced x-movements is more likely to succeed.

Conclusion

Our research aimed to develop multilinear regression models to predict the success rates of batters and pitchers based on various performance metrics. Through examination of different variables we've generated insight that may be valuable for player evaluation and performance improvement. The batting model suggests that improving launch angle sweet spot percentage, reducing swing length, and increasing exit velocity will result in a greater success rate for one pitch. Similarly, the pitching model identified the critical variables for a successful pitch by type.

Although our results are promising, the inability to analyze at-bat-level data, a standard practice in MLB evaluations, limits the robustness of our models. Future research can build upon this model by using such data and incorporating more players. Moreover, our models

that produce the ‘success-at-one-pitch’ percentage have potential to offer deeper insights into player capabilities and guide player development as well as game strategy.

Appendix

Below are the full data tables and screenshots of our batting model and pitching models.

OLS Regression Results						
Dep. Variable:	y	R-squared:	0.474			
Model:	OLS	Adj. R-squared:	0.416			
Method:	Least Squares	F-statistic:	8.115			
Date:	Sun, 28 Jul 2024	Prob (F-statistic):	0.000518			
Time:	18:14:07	Log-Likelihood:	-116.99			
No. Observations:	31	AIC:	242.0			
Df Residuals:	27	BIC:	247.7			
Df Model:	3					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	1.9331	16.274	0.119	0.906	-31.458	35.325
x1	0.5616	0.141	3.977	0.000	0.272	0.851
x2	-2.6716	1.502	-1.779	0.086	-5.753	0.409
x3	0.3162	0.173	1.829	0.078	-0.039	0.671
Omnibus:	1.559	Durbin-Watson:	2.041			
Prob(Omnibus):	0.459	Jarque-Bera (JB):	1.113			
Skew:	0.462	Prob(JB):	0.573			
Kurtosis:	2.905	Cond. No.	673.			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 2: Multilinear Regression Model for Batting

personId	num_at_bats	success_rate_data	success_rate_model
172804761	10	30.00	31.727216
174158975	14	15.38	8.747653
223971350	6	20.00	16.484367
290569727	9	44.44	33.520498
352830460	18	16.67	18.891913
360906992	14	7.14	16.636807
412098649	17	23.53	18.998611
432216743	5	20.00	9.391306
451871192	11	27.27	41.142537
459722179	19	0.00	19.262200
474808052	11	20.00	25.429779
485007791	20	20.00	27.915237
505414610	7	28.57	25.310467
518481551	4	25.00	26.912216
545569723	9	11.11	19.905343
558675411	11	36.36	21.840062
563942271	1	0.00	17.726546
568527038	8	12.50	16.613345
590082479	11	27.27	33.090484
617522563	5	40.00	22.725414
618024297	10	20.00	24.627046
654287703	7	57.14	46.542290
686425745	6	16.67	18.398412
719146721	18	22.22	16.802196
719210239	14	21.43	31.920569
765710437	12	54.55	38.918244
797796542	1	0.00	-3.739012
797957728	23	21.74	27.728642
805688901	9	22.22	31.122626
849653732	7	28.57	32.536633
854238128	9	55.56	28.210352

Table 4: Batting Data: Success Rates and Model Predictions ($n = 31$, $R^2 = 0.474$)

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=====
                        OLS Regression Results
=====
Dep. Variable:          y      R-squared:          0.015
Model:                  OLS    Adj. R-squared:       0.013
Method:                 Least Squares  F-statistic:      9.197
Date:                  Sun, 28 Jul 2024  Prob (F-statistic): 0.000108
Time:                  18:16:39  Log-Likelihood:   -878.75
No. Observations:      1233     AIC:             1764.
Df Residuals:          1230     BIC:             1779.
Df Model:              2
Covariance Type:       nonrobust
=====

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	coef	std err	t	P> t	[0.025	0.975]
const	0.5335	0.015	35.974	0.000	0.504	0.563
x1	0.0019	0.001	2.102	0.036	0.000	0.004
x2	0.0050	0.003	1.915	0.056	-0.000	0.010

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Omnibus:                4702.110  Durbin-Watson:       1.954
Prob(Omnibus):          0.000     Jarque-Bera (JB):    202.568
Skew:                   -0.186     Prob(JB):            1.03e-44
Kurtosis:               1.049     Cond. No.            21.2
=====

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Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 3: Multilinear Regression Model for All Pitches

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                        OLS Regression Results
=====
Dep. Variable:          y      R-squared:          0.037
Model:                  OLS    Adj. R-squared:       0.024
Method:                 Least Squares    F-statistic:      2.815
Date:                   Sun, 28 Jul 2024    Prob (F-statistic): 0.0976
Time:                   18:19:03    Log-Likelihood:   -53.078
No. Observations:       76    AIC:              110.2
Df Residuals:           74    BIC:              114.8
Df Model:                1
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	0.9911	0.260	3.816	0.000	0.474	1.509
x1	-0.0002	0.000	-1.678	0.098	-0.000	3.28e-05

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Omnibus:                 1197.468    Durbin-Watson:       2.029
Prob(Omnibus):           0.000    Jarque-Bera (JB):     11.116
Skew:                    -0.247    Prob(JB):             0.00386
Kurtosis:                1.193    Cond. No.             1.15e+04
=====

```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 1.15e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Figure 4: Multilinear Regression Model for Slider Pitches

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=====
                        OLS Regression Results
=====
Dep. Variable:          y      R-squared:          0.050
Model:                  OLS    Adj. R-squared:       0.041
Method:                 Least Squares    F-statistic:      5.478
Date:                   Sun, 28 Jul 2024    Prob (F-statistic): 0.0212
Time:                   18:19:33    Log-Likelihood:   -72.992
No. Observations:      106    AIC:              150.0
Df Residuals:          104    BIC:              155.3
Df Model:               1
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	-4.0792	1.989	-2.051	0.043	-8.024	-0.134
x1	0.0501	0.021	2.341	0.021	0.008	0.093

```

=====
Omnibus:                 1380.652    Durbin-Watson:       2.022
Prob(Omnibus):           0.000    Jarque-Bera (JB):     14.339
Skew:                    -0.267    Prob(JB):             0.000770
Kurtosis:                1.279    Cond. No.             3.92e+03
=====

```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 3.92e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Figure 5: Multilinear Regression Model for Sinker Pitches

OLS Regression Results						
=====						
Dep. Variable:	y	R-squared:				0.217
Model:	OLS	Adj. R-squared:				0.142
Method:	Least Squares	F-statistic:				2.903
Date:	Sun, 28 Jul 2024	Prob (F-statistic):				0.0771
Time:	18:19:58	Log-Likelihood:				-14.152
No. Observations:	24	AIC:				34.30
Df Residuals:	21	BIC:				37.84
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.3027	0.106	2.842	0.010	0.081	0.524
x1	-0.0353	0.019	-1.860	0.077	-0.075	0.004
x2	0.1204	0.058	2.062	0.052	-0.001	0.242
=====						
Omnibus:	3.843	Durbin-Watson:			2.018	
Prob(Omnibus):	0.146	Jarque-Bera (JB):			1.935	
Skew:	0.404	Prob(JB):			0.380	
Kurtosis:	1.868	Cond. No.			43.9	
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 6: Multilinear Regression Model for Cutter Pitches

OLS Regression Results						
=====						
Dep. Variable:	y	R-squared:		0.194		
Model:	OLS	Adj. R-squared:		0.137		
Method:	Least Squares	F-statistic:		3.375		
Date:	Sun, 28 Jul 2024	Prob (F-statistic):		0.0486		
Time:	18:20:16	Log-Likelihood:		-18.744		
No. Observations:	31	AIC:		43.49		
Df Residuals:	28	BIC:		47.79		
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-8.7082	5.154	-1.690	0.102	-19.265	1.849
x1	0.1152	0.061	1.894	0.069	-0.009	0.240
x2	-0.1694	0.091	-1.860	0.073	-0.356	0.017
=====						
Omnibus:		9.241	Durbin-Watson:		2.472	
Prob(Omnibus):		0.010	Jarque-Bera (JB):		2.573	
Skew:		-0.248	Prob(JB):		0.276	
Kurtosis:		1.679	Cond. No.		5.23e+03	
=====						

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
 [2] The condition number is large, 5.23e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Figure 7: Multilinear Regression Model for Changeup Pitches